

F015 Final Integrated Report

Final Report: Questionnaire Embeddings Phase 1–4 Integrated Analysis

Scope and deliverables

This F015 report aggregates existing accepted artifacts only. It does not rerun Phase 1–4 experiments, regenerate embeddings, or re-estimate bootstrap tests.

Primary metric: **item-level Pearson r** . Key secondary metrics: **trait_r_mean**, **profile_r**, **MAE**.

Executive summary

Phase 1 selected **Coverage** as the strongest item-selection strategy in the observed Table 1 cells (best cell $m=90$, $\text{item}_r=0.4842$).

Phase 2 selected **SoftmaxKNN** as the recommended predictor family over Coverage-selected items.

Phase 3 diagnostics suggested **E5-base-v2** for highest selected-set coverage and **MPNet-base-v2** for strongest within-minus-between trait separation.

Phase 4 best observed pipeline cell was **A2_tuned / BGE-base-en-v1.5** at $m=90$, $\text{item}_r=0.6167$.

Largest positive B–A re-selection contribution was **B1_fixed minus A1_fixed** for MPNet-base-v2 at $m=10$, $\Delta \text{item}_r=0.0622$.

Phase 1: item-selection contribution

Phase 1 compared random, balanced-random, semantic Coverage, Coverage+Diversity, TraitPredictiveness, and Hybrid strategies. The final report carries forward Coverage as the preferred semantic selection strategy and treats Phase 1 Table 1/Table 2 as the source of item-level and trait-level evidence.

Phase 2: prediction-algorithm contribution

Phase 2 compared Tuned UniformKNN, CosineWeightedKNN, SoftmaxKNN, KernelSmoothing, and the cross-phase original-paper UniformKNN $K=5$ baseline. The final Phase 4 pipeline uses SoftmaxKNN because it gave the best overall item-level Pearson r in Phase 2 and supports both fixed Phase 2 hyperparameters and train-inner tuning.

Phase 3: embedding-space diagnostics

Phase 3 compared SBERT original, MiniLM, MPNet, E5, and BGE using Coverage/Redundancy and raw-cosine trait-structure diagnostics. These diagnostics are explanatory hypotheses, not predictive-performance claims; Phase 4 is the predictive test.

Phase 4: A1/A2/B1/B2 attribution

****A1_fixed**** = fixed historical SBERT Coverage `S\_old` plus fixed Phase 2 SoftmaxKNN hyperparameters. This isolates embedding-neighbor geometry under a fixed administered item set.

Phase 4 attribution, reproducibility, and limitations

****A2_tuned**** = fixed ``S_old`` plus embedding-specific train-inner K/τ tuning. This estimates geometry contribution after calibration without changing item selection.

****B1_fixed**** = embedding-specific Coverage re-selection ``S_new`` plus fixed Phase 2 hyperparameters. This adds re-selection contribution while keeping hyperparameters fixed.

****B2_tuned**** = embedding-specific ``S_new`` plus embedding-specific train-inner K/τ tuning. This is the full re-selected+tuned Phase 4 pipeline.

The B–A comparisons are paired within the same embedding and tuning regime, so B1–A1 and B2–A2 quantify selection contribution beyond the corresponding fixed-``S_old`` baseline.

Best pipeline by administered-item ratio

Statistical inference

The primary Phase 4 statistical comparisons use paired bootstrap over participants while preserving outer-fold pairing. New embeddings are compared against ``sbert_original``; B–A selection-contribution tests compare B1 vs A1 and B2 vs A2 within the same embedding and ratio. Raw p-values plus Holm and Benjamini-Hochberg corrected p-values are preserved in ``statistical_summary.csv``.

Reproducibility notes

Random state is fixed at 0 in upstream CV and selection code.

Outer folds are participant-level folds and are paired across Phase 4 A/B comparisons.

A2/B2 K and τ values are selected only on train-inner validation splits; test participants are held out for final evaluation.

Phase 4 primary predictions are continuous and clipped to [1,5] without rounding; rounded accuracy and rounded MAE are supplemental outputs only.

This report reads existing CSV/TXT/PNG artifacts and records source hashes in ``report_manifest.json``.

Output inventory

``final_summary.csv``: normalized Phase 1–4 and recommended-pipeline rows.

`statistical\_summary.csv`: normalized inferential tests from Phase 1, Phase 2, and Phase 4.

`best\_pipeline\_table.csv`: best observed Phase 4 pipeline by 10/30/50/90 administered items.

`figures/table5\_phase4\_integrated\_synthesis.csv` (**Table 5**): A-vs-B synthesis table for Phase 4.

`figures/figure6\_phase4\_selection\_vs\_performance.pdf/png` (**Figure 6**): new integrated selection/performance synthesis figure.

Limitations

Current conclusions are primarily based on NEO-PI-R; cross-questionnaire generalization remains to be tested in future work.

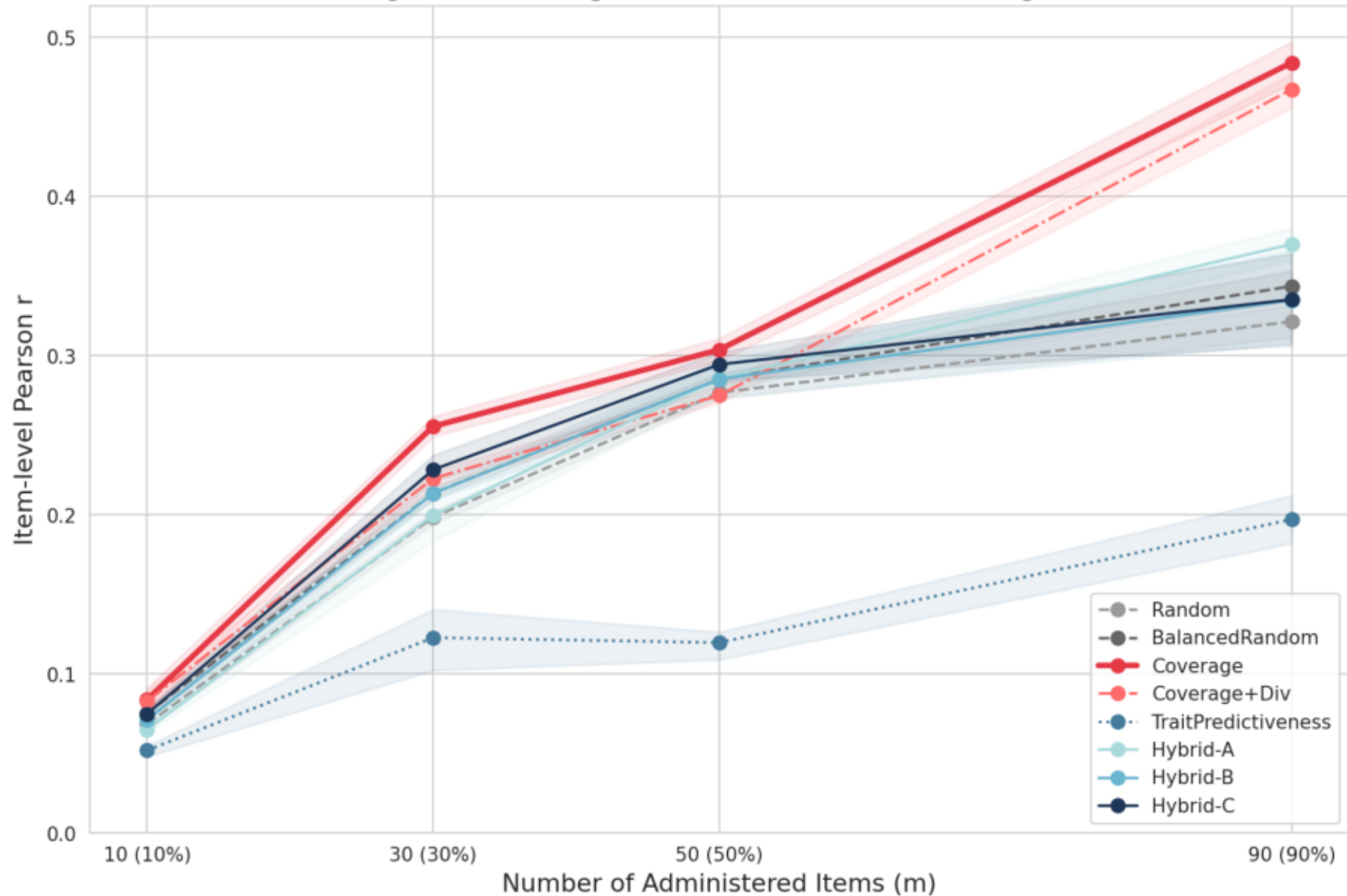
No cross-questionnaire generalization experiment is added in F015; the report is an aggregation and synthesis of existing NEO-PI-R artifacts.

Best pipeline by administered-item ratio

ratio	recommended_version	embedding_label	selection_scope	K_mode	tau_mean	item_r_mean	trait_r_mean	profile_r_mean	item_mae_mean
10	B1_fixed	MPNet-base-v2	bedding_specific_	7	0.1000	0.3507	0.6776	0.7715	0.9836
30	A2_tuned	MPNet-base-v2	S_old_fixed	3	0.1000	0.4474	0.8884	0.9002	0.8961
50	A2_tuned	MPNet-base-v2	S_old_fixed	5	0.1000	0.4896	0.9452	0.9486	0.8442
90	A2_tuned	BGE-base-en-v1.5	S_old_fixed	3	0.1200	0.6167	0.9943	0.9943	0.6946

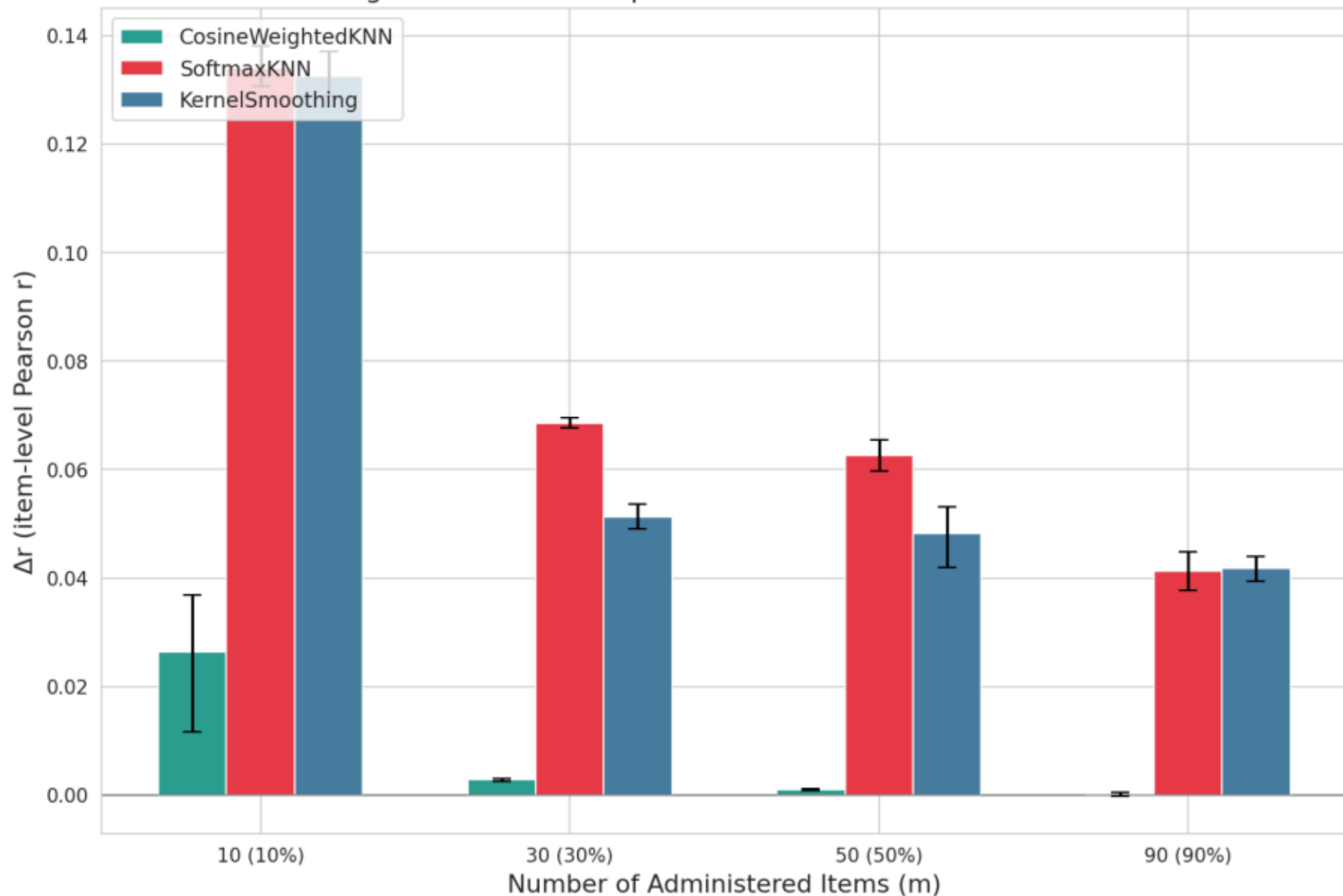
Phase 1 learning curve

Figure 1: Learning Curve — Item Selection Strategies



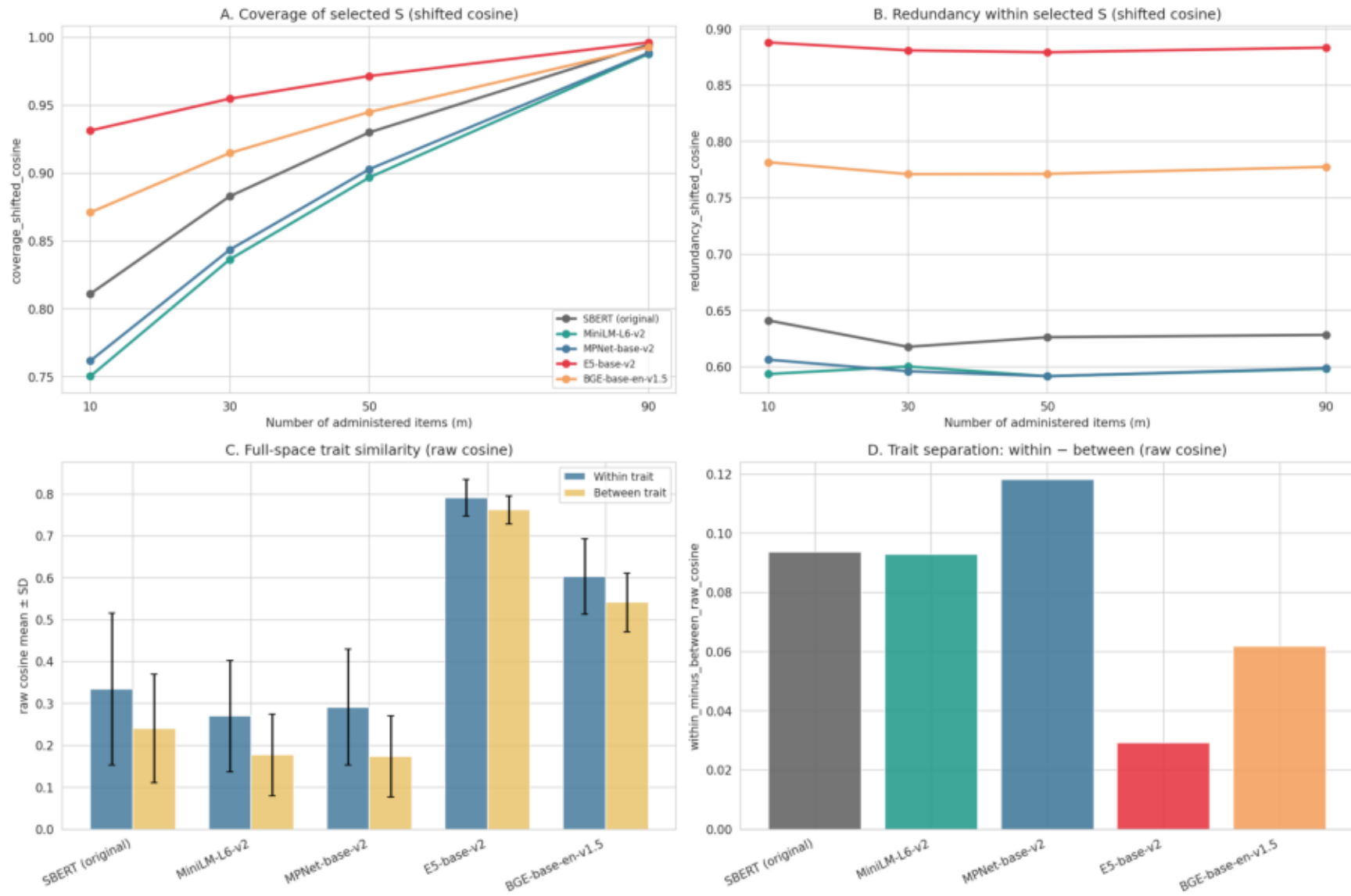
Phase 2 predictor deltas

Figure 3: Predictor Improvement over Tuned UniformKNN



Phase 3 embedding diagnostics

Figure 5: Embedding Space Quality Diagnostics



Phase 4 A/B learning curves

Figure 4 — Phase 4 A/B embedding learning curves

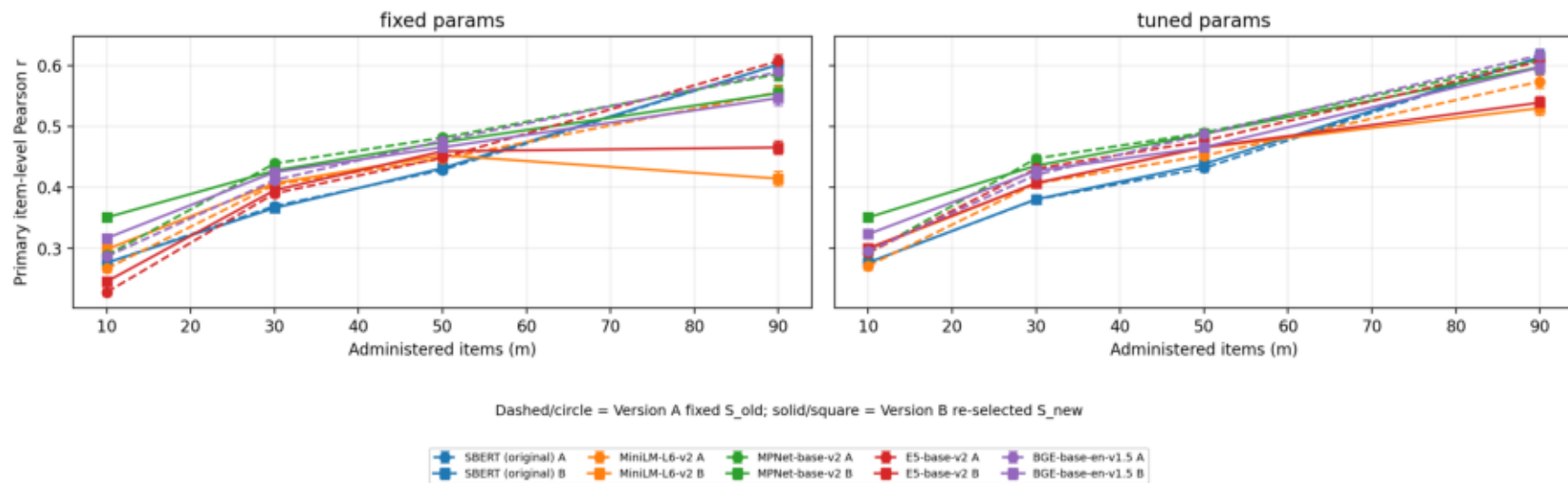


Figure 6 — new integrated Phase 4 synthesis

Figure 6 — Integrated Phase 4 selection/performance synthesis

